

GMRTT35D Rear Helper Spring Kit Installation Manual

Model: GMRTT35D (Rear Suspension Package)

Please read these instructions carefully before starting the installation. Ensure all standard safety precautions are observed while working underneath the vehicle.

1. Parts List

Before beginning, verify that all components listed below are present in your kit.

Ref #	Description / Specifications	Qty
1	Hex Cap Screw (M10 x 80 x 1.5, Grade 8.8, Zinc Plated)	2
2	Locator (3.36" Outer Diameter x 1" Thick)	2
3	Rubber Spacer (4" Outer Diameter x 1" Thick, with Bushing)	2
4	Hollow Rubber Spring	2
5	Fender Washer (3/8" x 1-1/2", Grade 2, Zinc Plated)	2
6	Nylock Locknut (M10 x 1.5, Grade 8.0, Zinc Plated)	2

2. Installation Instructions

- 1. Raise Vehicle:** Elevate the vehicle using an appropriate hoist or jack, allowing the axle to hang freely. Secure the chassis with appropriate stands before proceeding.
- 2. Remove OEM Bump Stop:** Detach the existing OEM vehicle bump stop by prying it from its mounting cup.
- 3. Position Spring Assembly:** Insert the pre-assembled helper spring assembly into the OEM bump stop cup.
- 4. Lubricate and Seat Spring:** Apply a soapy water solution to the helper spring assembly. Position it into the bump stop cup, applying simultaneous pushing and twisting force to fully seat it.
- 5. Verify Assembly Seating:** Confirm that the helper spring assembly is correctly seated within the bump stop cup.
- 6. Ensure Final Seating:** To ensure proper seating, temporarily place a short 2x4 wood block between the axle and the helper spring assembly. Lower the vehicle onto the block, then raise the vehicle again and remove the block.
- 7. Lower Vehicle:** Carefully lower the vehicle to the ground.

Installation Notes:

- The helper spring kit is supplied pre-assembled. Install the assembly as supplied for standard applications.

3. Clearance Check & Fitting Guidance

- Clearance Check Reference:** With the vehicle unloaded and at normal ride height, approximately **1/2" to 1-1/2" (13-38 mm)** may be used as a reference range when checking clearance between the rubber spring and its contact point. For factory-height applications, about 1" (25 mm) is a typical reference within this range.
- Larger Gap:** A leveling kit, suspension lift, or other ride-height modification may increase the gap. A larger gap may delay helper spring engagement and reduce support.
- Smaller Gap:** Too little clearance may cause earlier spring engagement and may result in a firmer unloaded ride.
- Under Load:** Cargo, accessories, or vehicle load may reduce or eliminate the gap as the helper spring engages.
- Installation Check:** Actual clearance may vary with vehicle configuration, ride height, suspension condition, and load. If the unloaded gap is significantly outside this reference range, verify ride height, suspension modifications, load condition, and correct installation.

Important Safety Warnings:

- Do not exceed the vehicle manufacturer's recommended axle weight rating / GAWR.
- Inspect for potential interference between the hollow rubber spring and adjacent vehicle components. Under load, the rubber spring will expand in diameter.